

Depth of Vocabulary Knowledge

STUART WEBB

Depth of vocabulary knowledge refers to how well words are known. Developing vocabulary depth typically involves the accumulation of knowledge through encountering and using words in a variety of different contexts in order to learn the forms, meanings, and uses of words. Knowledge of a word's spelling, pronunciation, derivations and inflections, meaning senses, semantic associations, collocations, and grammatical functions, and when it may be appropriate or inappropriate to use it are all required to fully know words (Nation, 2001). Vocabulary depth is indicated by the extent to which these aspects of vocabulary knowledge are present, and this in turn signals the extent to which words may or may not be used successfully.

The term "depth of knowledge" was introduced as one dimension of vocabulary knowledge by Anderson and Freebody (1981) in contrast to a second dimension, breadth of knowledge. Breadth refers to how many words are known and is typically indicated by scores on tests of vocabulary size, such as the Eurocentres Vocabulary Size Test (Meara & Jones, 1990) and the Vocabulary Size Test (Nation & Beglar, 2007). Breadth is usually determined by the degree to which learners are able to link form to meaning. Being able to link form to meaning is an important step in vocabulary development because it increases the likelihood that words may be understood and used. However, it does not ensure that these words will be understood and used correctly and this is why there is also a need for learners to develop vocabulary depth. Several studies have found positive correlations between depth and breadth of knowledge (Nurweni & Read, 1999; Qian, 1999; Vermeer, 2001). However, in each of these studies there was partial overlap between the constructs measured by the tests making further research warranted.

It is important for teachers to consider depth when planning the vocabulary learning component of a course because accumulating the knowledge necessary to understand and use words fluently may often be a long process. Awareness of how different types of learning may affect vocabulary knowledge can help teachers to design and sequence activities to promote depth. Teachers may also look at depth of knowledge when diagnosing language weaknesses. Measuring depth of knowledge can show us what learners know and do not know about words. For example, unusual word combinations in speech and writing often signal limitations in knowledge of collocation, while an inability to produce different derived forms of a word may indicate a lack of knowledge of word parts. Awareness of vocabulary depth can also help teachers to see how tasks and activities contribute to learning and the need for different types of tasks to be used together to develop richer knowledge of words. It is also essential for researchers to take vocabulary depth into consideration when measuring learning. Measuring learning with a single test of form and meaning may only measure a fraction of what may be known about a word.

What Is Depth of Knowledge?

There have been many descriptions of vocabulary knowledge (Richards, 1976; McCarthy, 1990; Nation, 1990, 2001; Aitchison, 1994; Laufer, 1997; Henriksen, 1999; Miller, 1999;

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Table 1 Description of vocabulary knowledge (from Nation, 2001, p. 27)

Form	Spoken	R	What does the word sound like?
		P	How is the word pronounced?
	Written	R	What does the word look like?
		P	How is the word written and spelled?
	Word parts	R	What parts are recognizable in this word?
		P	What word parts are needed to express the meaning?
Meaning	Form and meaning	R	What meaning does this word form signal?
		P	What word form can be used to express this meaning?
	Concept and referents	R	What is included in the concept?
		P	What items can the concept refer to?
	Association	R	What other words does this make us think of?
		P	What other words could we use instead of this one?
Use	Grammatical functions	R	In what patterns does this word occur?
		P	In what patterns must we use this word?
	Collocation	R	What words or types of words occur with this one?
		P	What words or types of words must we use with this one?
	Constraints on use	R	When, where, and how often would we expect to meet this word?
		P	Where, when, and how often can we use this word?

Note. R = receptive knowledge, P = productive knowledge.

Schmitt, 2000; Read, 2004). However, there is no definition of vocabulary depth that is widely agreed upon. Nation's (2001) description of what is involved in knowing a word is the most comprehensive account of depth. He classifies vocabulary knowledge according to form, meaning, and use; each category contains three aspects of knowledge and each aspect of knowledge can be broken down into receptive and productive components. This results in 18 unique components of vocabulary knowledge. Vocabulary depth is the extent to which each of these components is present. Table 1 shows Nation's description of vocabulary knowledge.

Receptive or passive knowledge is what is required to understand words when they are encountered in listening and reading. Productive or active knowledge is the knowledge necessary to use a word in speech and writing. For example, receptive knowledge of written form involves recognizing the spelling of a word when it is encountered and receptive knowledge of form and meaning involves recognizing the meaning that a particular form conveys. On the other hand, productive knowledge of written form involves being able to spell a word correctly and productive knowledge of form and meaning involves being able to produce a word that conveys the intended meaning.

There is often an assumption that receptive knowledge precedes productive knowledge. However, this may only be true if we look at the individual aspects of vocabulary knowledge rather than the broader construct of knowing words. For example, the ability to recognize the spelling of the written form of a word is likely to precede the ability to spell it correctly, or the ability to recognize that a specific written form conveys a specific meaning is likely to precede the ability to convey a specific meaning using the correct word form. In contrast, it may be possible to spell a word correctly (productive knowledge of written form) or use the word with grammatical accuracy (productive knowledge of grammatical functions) before the meaning of the word is understood (receptive knowledge of form

and meaning). Thus, to examine receptive and productive vocabulary knowledge, it is necessary to differentiate between receptive and productive knowledge of individual aspects of knowledge rather than receptive and productive knowledge of words as a whole.

When considering vocabulary depth, it is important to differentiate between strength of knowledge and depth of knowledge (Nation & Webb, 2011). Strength of knowledge refers to how well a single aspect is known. Each of the aspects can be learned to different degrees, from no knowledge to varying degrees of partial knowledge to full knowledge. For example, being able to identify the correct spelling of a word (receptive knowledge of written form) from among a number of unrelated spellings may indicate minimal knowledge of written form. Identification of the correct spelling of the word from among a number of closely related spellings may indicate greater knowledge of written form, and correctly spelling the word may indicate full knowledge of written form. Strength of knowledge may move in both directions. Knowledge is likely to increase the more a word is encountered and used, but if there are long gaps between encounters, knowledge may decrease. Strength of knowledge provides some indication of depth of knowledge. However, because measuring strength of one aspect involves assessing only a portion of the construct of depth, researchers should be careful not to generalize from strength to depth. Gaining depth of knowledge involves learning all nine aspects of knowledge.

Measuring Depth of Knowledge

Research on assessing vocabulary depth is still in its infancy. Read (2004) reports that depth has been operationalized in research in three ways: precision of meaning, comprehensive word knowledge, and network knowledge. These categories provide some direction on how depth might be measured. However, each one is limited in the degree to which depth is measured, and overlap between the categories suggests that together they may still not provide an accurate assessment of depth.

The Vocabulary Knowledge Scale (VKS: Wesche & Paribakht, 1996) and the Word Associates Test (WAT: Read, 1993, 1998) are established tests designed with the purpose of measuring vocabulary depth. Both tests were useful improvements on traditional measures of form and meaning. However, both are limited in the degree to which they indicate depth. The VKS uses the following scale to measure depth:

1. I don't remember having seen this word before.
2. I have seen this word before, but I don't know what it means.
3. I have seen this word before, and I think it means _____.
4. I know this word. It means _____.
5. I can use this word in the sentence: _____.

Although different points on the scale may indicate an increase in knowledge, it is impossible to tell if there is an ordinal progression in any aspect of knowledge besides form and meaning. For example, at only point 5 on the scale is grammatical accuracy required. However, knowledge of the grammatical functions of an item may be present at a lower point on the scale; it is possible to use a word correctly in a sentence without knowing its meaning.

The WAT measures knowledge of three aspects—form and meaning, concept and referents, and collocation—and provides an overall score indicating how well items are known. However, it does not provide a separate measure of each of these aspects, so it is not clear to what extent each aspect is known. For example, one testee could score solely on their knowledge of form and meaning for all items, and another could score solely on

their knowledge of collocation. The test would show that both testees had a similar amount of knowledge and lacked some degree of depth (a useful finding) but would not differentiate between what was known and what was lacking. Moreover, if the known aspect was always form and meaning then the test would have little value as a measure of depth. Separate tests focusing on individual aspects may provide a more accurate measurement of depth.

A more useful approach to measuring depth is to examine multiple aspects of knowledge and measure the strength of knowledge of each of those aspects. A series of studies by Norbert Schmitt and his colleagues (Schmitt & Meara, 1997; Schmitt 1998, 1999; Pigada & Schmitt, 2006) and Webb (2005, 2007a, 2007b, 2008, 2009a, 2009b) have used this approach. Schmitt's (1998) study was particularly innovative as he used interviews to measure different degrees of strength of knowledge of written form, associations, grammatical information, as well as form and meaning. By measuring strength of each aspect of knowledge he was able to show that there is movement in multiple aspects of knowledge over time; thus knowledge of an aspect is not a dichotomy (complete or incomplete) but tends to gradually increase or decrease in degrees of knowledge. Webb (2005, 2007a, 2007b, 2009a, 2009b) examined the effects of different vocabulary-learning activities on five aspects of knowledge. He created receptive and productive tests to isolate and measure written form, form and meaning, association, collocation, and grammatical functions. In a study examining the effects of repetition on vocabulary knowledge, Webb (2007b) showed the value of measuring different aspects of knowledge to evaluate learning. As the number of repetitions increased, vocabulary knowledge increased to different degrees; measuring multiple aspects indicated significantly greater knowledge of some aspects but not others. The results would not always have indicated differences in knowledge if only a single aspect had been assessed. The research also highlighted the need for multiple measures at different sensitivities to measure strength of knowledge of each aspect. Often there was no difference in knowledge between groups at one level of sensitivity (receptive or productive), but there was a difference at the other level. This finding builds on Nagy, Herman, and Anderson's (1985) research, which demonstrated the importance of using multiple tests at different sensitivities to provide an accurate assessment of form and meaning.

Meara and Wolter (2004) argue for a different approach to measuring depth. They report that using a battery of tests to measure depth for a small number of words is impractical, and instead suggest measuring depth for sets of words in a similar way to how vocabulary size is examined. They use the V_Links software version 2.00 to measure knowledge of associations for sets of 20 words taken from the most frequent 1,000 words. The study provides a useful approach to how depth and vocabulary size could be measured individually to provide a more comprehensive measurement of vocabulary knowledge.

Developing Vocabulary Depth

Planning vocabulary learning is essential if learners are to develop vocabulary depth. Teachers must decide which words are to be learned during a course and how they will provide the opportunities necessary to develop knowledge of those words (Nation, 2008). A lack of planning may lead to an inability of students to successfully use the words that they have learned—a situation common to many foreign-language classrooms. Nation's (2001, 2007) four strands (meaning-focused input, meaning-focused output, language-focused learning, fluency development) provide a useful approach to developing vocabulary depth. Each strand may contribute to vocabulary development in different ways, and allocating a similar amount of time to learning in each strand is likely to lead to deeper learning than learning solely in one strand.

Language-focused learning includes deliberate learning of new words. Although activities can be created and modified to focus on multiple aspects of vocabulary knowledge, many activities focus on strengthening the link between form and meaning. This is a logical place to start with learning. The other strands can build upon this initial learning of form and meaning to develop the comprehensive vocabulary knowledge necessary to use and understand words fluently.

In the meaning-focused input strand, words should be encountered repeatedly in a variety of original contexts. It is well established that repeated encounters in context contribute to learning form and meaning but are also likely to help develop all other aspects of knowledge (Pigada & Schmitt, 2006; Webb, 2007b) with the mode of input (spoken or written) having an effect on the degree to which spoken and written form is learned. Meaning-focused input is likely to have a greater effect on increasing receptive knowledge than on productive knowledge.

Meaning-focused output involves using language in speech and writing. The meaning-focused output strand may help to push vocabulary knowledge from receptive understanding to productive use. The fluency development strand probably receives the least attention in courses. However, it also plays an important role in facilitating vocabulary depth. In this strand, the aim is to use or understand words in a native-like manner. Time pressure is one of the criteria to help facilitate increased processing speed and vocabulary fluency. It may be that this move toward native-like lexical access speed in each of the four skills may indicate the degree of word mastery.

Few teaching techniques are designed to develop all aspects of knowledge, and most are focused on strengthening the link between form and meaning. For example, to successfully complete matching activities, multiple-choice questions, and crossword puzzles, students typically need to correctly link form and meaning. There is justification for focusing on form and meaning because it allows the meaning of words encountered to be understood. Also, if there is overlap in L1 and L2 vocabulary knowledge, learners may gain knowledge of other aspects by linking L1 meaning to L2 form (Webb, 2007c). One aspect of knowledge that receives little attention in vocabulary-learning activities but deserves more attention is word parts. Learning word parts such as the prefixes *mid-*, *mis-*, and *multi-* and the suffixes *-able*, *-ee*, and *-er* provides knowledge that can be used to aid future vocabulary learning. For example, knowing that *mid-* in *midnight* means 'middle' and that *-ee* in *employee* means 'person' may help learners to understand other items that contain the same word parts when they are encountered, for example in *midsummer*, *midlife*, *referee*, and *nominee*.

The order of learning the aspects of knowledge may depend on how a word is learned. There is some evidence that, when words are encountered during reading, knowledge of written form is gained first (Pigada & Schmitt, 2006; Webb, 2007b). Schmitt (2010) suggests that typically the written and spoken form are most likely learned first followed by form and meaning and grammatical functions, while the aspects that relate to use may be learned last.

Students need to be aware of what is involved in knowing a word if teachers expect them to effectively learn words. Teachers can raise awareness of depth of vocabulary knowledge in a number of ways. Activities such as Schmitt and Schmitt's (1995) vocabulary notebooks, which explicitly focus attention on different aspects of knowledge, are useful for this. Concordances are also a useful tool for developing depth of knowledge. A concordance is a collection of examples of the use of a word. By seeing a word used in many different ways, knowledge of form, meaning, and use may be gained. Another useful approach is to use a sequence of activities that each focus on developing different aspects of knowledge. Sökmen (1992) provides an excellent example of how this might be done. She describes an approach which begins with students learning the form and meaning of unknown words. This is followed with activities that each focus on increasing

knowledge of another aspect of knowledge for those words until finally students may have gained knowledge of form and meaning, spoken and written form, word parts, associations, collocations, grammatical functions, and concept and referents.

Developing the students' vocabulary depth should also include measuring their depth of knowledge. This provides teachers with a more complete picture of the students' knowledge and also demonstrates to students the importance of learning other aspects as well as form and meaning. Tests and quizzes can vary the aspects of knowledge that are examined. For example, one quiz might measure knowledge of form and meaning and another might measure knowledge of collocation. The questions within a quiz might also be written to measure different aspects of knowledge. The following questions show one way in which different aspects of knowledge might be measured for a word.

Underline the word that is not likely to be used together with *accept* (receptive knowledge of collocation):

1. offer
2. holidays
3. invitation
4. responsibility

Underline the correct spelling (receptive knowledge of written form):

1. acapt
2. acsept
3. accept
4. axept

Underline the incorrect derivation (receptive knowledge of word parts):

1. acception
2. accepting
3. acceptance
4. acceptable

Underline a word with an unrelated meaning (receptive knowledge of association):

1. decline
2. receive
3. refuse
4. reduce

Underline the correct sentence (receptive knowledge of grammatical functions):

1. She accepts.
2. It is accept.
3. It is an accept.

Conclusion

As we have seen, it is important for teachers, learners, and researchers to have an understanding of vocabulary depth. Developing depth of knowledge should influence how courses with a vocabulary-learning component are designed and how learners approach their vocabulary learning. Awareness of what is involved in knowing a word can highlight what aspects of knowledge need to be measured to help researchers to more accurately assess learning.

SEE ALSO: Assessment of Vocabulary; Explicit Learning in Second Language Acquisition; Incidental Learning in Second Language Acquisition; Lexical Collocations; Teaching Vocabulary; Vocabulary Acquisition in Second Language Acquisition; Vocabulary Size in a Second Language; Vocabulary Size in the First Language

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Suggested Reading

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